

RESEARCH ARTICLE SUMMARY

DIRECTED EVOLUTION

Rapid in silico directed evolution by a protein language model with EVOLVEpro

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INTRODUCTION: Protein engineering is a critical tool in basic biology and medicine, but current methods have multiple shortcomings. Directed evolution is labor intensive, can get trapped in local maxima, and is not compatible with complex objectives. In silico optimization with protein language models (PLMs) has some power to predict functional mutations but often fails to substantially improve function. Emerging generative methods have shown progress in engineering simple objectives, such as binding, but cannot generalize to improve diverse protein functions. We introduce EVOLVEpro, a protein engineering method that combines PLMs with few-shot active learning on a regression model to augment experimental directed evolution. EVOLVEpro

rapidly improves protein activity with sparse experimental data, overcoming limitations of current directed evolution and in silico protein engineering approaches.

RATIONALE: Protein language models, trained across broad sequence diversity, learn emergent properties from natural protein evolution, but the relationship between model representation and protein activities is unclear. We hypothesized that iterative experiments could rapidly elucidate the relationship between PLM embeddings and desired protein functions or attributes, enabling directed optimization of proteins towards desired goals. EVOLVEpro interprets PLMs embeddings through a top-layer regression model, learning the relation-

ship between sequence and experimentally determined activities through an iterative active learning process. The lightweight random forest regression model can optimize multiple protein properties simultaneously during iterative rounds of testing with as few as 10 experimental data points per round, enabling complex multiobjective evolution campaigns and minimal experimental setup.

RESULTS: EVOLVEpro outperformed zero-shot methods in benchmarks across 12 deep mutational scanning datasets, including epitope binding, nucleic acid binding, and enzyme catalysis. We used EVOLVEpro to engineer six different proteins with diverse applications. We improved the binding affinity two monoclonal antibodies by up to 40-fold, the indel formation activity of a miniature CRISPR nuclease by fivefold, the insertion efficiency of a prime editor by twofold, the integration efficiency of a serine integrase by fourfold, and the transcription fidelity and mRNA quality of a T7 RNA polymerase by 100-fold. We found that EVOLVEpro predictions are substantially different and superior to predictions by the base layer alone, underscoring the importance of PLM interpretation by regressors and experimental data. EVOLVEpro-nominated mutations are distinctive from an evolutionary perspective, insightful from a structural analysis perspective, and based on iteration through position-specific hypotheses explored across rounds of evolution.

CONCLUSION: With high success rates at improving protein function, EVOLVEpro is advantageous relative to existing protein engineering methods. The model requires minimal initial knowledge of the target protein and can optimize over multiple objectives that would otherwise be inaccessible to typical directed evolution approaches. These features make EVOLVEpro well-suited for the engineering of challenging proteins for both basic biology and translational applications. The lack of correlation between base PLM and EVOLVEpro predictions demonstrates the need for interpretation of foundational models through the lens of experimentation and suggests that further hybrid approaches combining in silico and experimental methods will increase our capacity to engineer and understand protein function landscapes across the diversity of life. ■

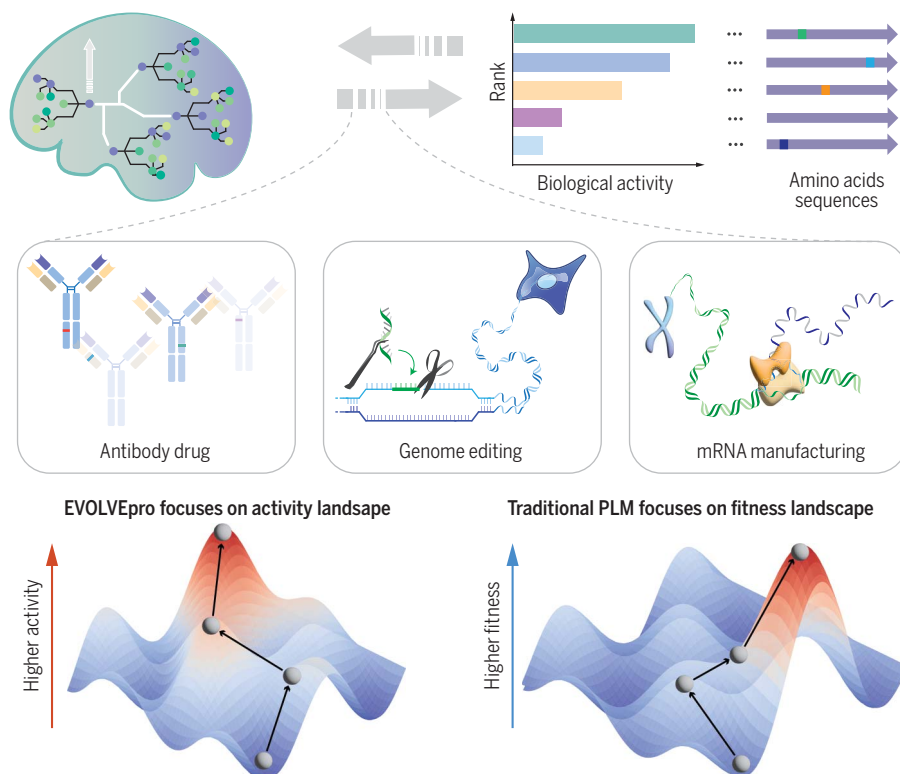
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EVOLVEpro learns activity landscape for rapid in silico protein engineering. EVOLVEpro explores functional landscapes rather than fitness landscapes, engineering diverse proteins, such as monoclonal antibodies, genome editing machinery, and mRNA-producing enzymes, to high levels of activity.

RESEARCH ARTICLE

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Rapid in silico directed evolution by a protein language model with EVOLVEpro

Kaiyi Jiang^{1,2,3,4,†}, Zhaoqing Yan^{1,2,3,†}, Matteo Di Bernardo^{5,†}, Samantha R. Sgrizzi^{1,2,3}, Lukas Villiger⁶, Alisan Kayabolen^{1,2,3}, B. J. Kim⁷, Josephine K. Carscadden^{1,2,3}, Masahiro Hiraizumi⁸, Hiroshi Nishimasu^{8,9,10}, Jonathan S. Gootenberg^{1,2,3,*†}, Omar O. Abudayyeh^{1,2,3,*†}

Directed protein evolution is central to biomedical applications but faces challenges such as experimental complexity, inefficient multiproperty optimization, and local maxima traps. Although in silico methods that use protein language models (PLMs) can provide modeled fitness landscape guidance, they struggle to generalize across diverse protein families and map to protein activity. We present EVOLVEpro, a few-shot active learning framework that combines PLMs and regression models to rapidly improve protein activity. EVOLVEpro surpasses current methods, yielding up to 100-fold improvements in desired properties. We demonstrate its effectiveness across six proteins in RNA production, genome editing, and antibody binding applications. These results highlight the advantages of few-shot active learning with minimal experimental data over zero-shot predictions. EVOLVEpro opens new possibilities for artificial intelligence-guided protein engineering in biology and medicine.

Protein diversity has been shaped by billions of years of evolutionary pressure, filtering the potential design space for diverse biological functions. Understanding the connection between protein amino acid sequence and function is crucial for advancing biology and developing new therapeutics. Experimental approaches, such as deep mutational scanning (DMS), can directly measure functional effects of protein mutations (1–3) but are limited to exploring only a fraction of the possible protein sequence space. Computational approaches leveraging orthologous sequences can reduce the experimental data needed to map fitness landscapes and optimize proteins (4–6), but these methods require multiple sequence alignments, high ortholog abundance, and minimal sequence length variation.

To overcome these challenges, fitness can be inferred by training broadly across evolutionary diversity. Protein language models (PLMs), such as ESM-2 (7), are trained across comprehensive protein sequence databases to fill in missing amino acids. PLMs learn informative biological representations (7–12), protein structure prediction (7), and functional annotation (13). PLMs have nominated protein mutants with improved activity (14, 15), and generative PLMs (16–18) have been used to design newly discovered proteins. However, zero-shot-predicted mutants have limited success (14, 15), and de novo-designed proteins typically exhibit lower or comparable activity relative to natural wild-type (WT) sequences (16, 19). Although zero-shot models can predict antibody mutations to increase binding affinity, they cannot improve other important antibody features, such as developability and immunogenicity (15, 20). These failures of PLMs to substantially improve protein activity in zero-shot settings are driven by their inability to generalize to new contexts owing to limited training data (21) and the difference between evolutionary fitness and protein function. Therefore, protein optimization and interpretation by using PLM-based approaches require additional experimental data to reach design-specific objectives.

Iterative approaches for optimization, such as directed evolution (DE) (22), take advantage of the smoothness of protein fitness landscapes to improve function. Although these methods are successful in contexts with suitable activity landscapes and screening methods, they can be labor intensive and fail on rugged landscapes, especially when trapped in local optima. Some improvements can be gained from com-

binning DE with machine learning. Machine learning-directed evolution (MLDE) methods (23–28) leveraging active learning have effectively improved diverse proteins but at the cost of comprehensive experimental evaluation. Merging active learning with simpler protein representation models, such as recurrent neural networks (29, 30), has simplified the evolution process, but previous attempts at active learning on protein models (30) have not generalized well beyond proof-of-concept demonstrations, such as fluorescent protein engineering, owing to shortcomings in the protein representation space.

Given the limited success of PLM-based methods to rank variant effects in a zero-shot fashion (15) or to iteratively nominate protein mutations (30), we reasoned that active learning with high-performance PLMs and small sets of mutants would improve directed evolution performance. In this study, we present the protein evolution model EVOLVEpro (evolution via language model-guided variance exploration for proteins) to solve these challenges with MLDE. EVOLVEpro nominates high-activity protein variants with active learning and minimal experimental testing, achieving rapid prediction of high-activity mutants. This performance stems from a modular approach that marries an evolutionary-scale PLM with a top-layer regression model to learn a protein's activity landscape and guide the directed evolution process in silico. This top-layer model is trained over multiple rounds of evolution, with each round evaluating a small set of EVOLVEpro-predicted mutants by using one or more experimental assays. These data are then used to update the model and predict the next round of mutation candidates. We hypothesize that combining the PLM and regressor enables the identification of multiple high-activity regions simultaneously, providing generalization across diverse protein classes with rugged activity landscapes and robustness to local optima due to detrimental or neutral mutants (28). Moreover, EVOLVEpro prompting only uses protein sequences and does not require structural information, expert knowledge, or prior data. We demonstrate EVOLVEpro's ability to evolve multiple activities of a protein simultaneously, opening up vast possibilities for its use in biology and medicine.

Development and benchmarking of the EVOLVEpro model

We developed a deep learning-based directed evolution framework, EVOLVEpro, combining (i) a PLM to encode protein sequences into a continuous latent space to facilitate activity optimization and (ii) a top-layer regression model to learn the mapping between latent space and activity from a few number of data points (i.e., the low-N regime). EVOLVEpro

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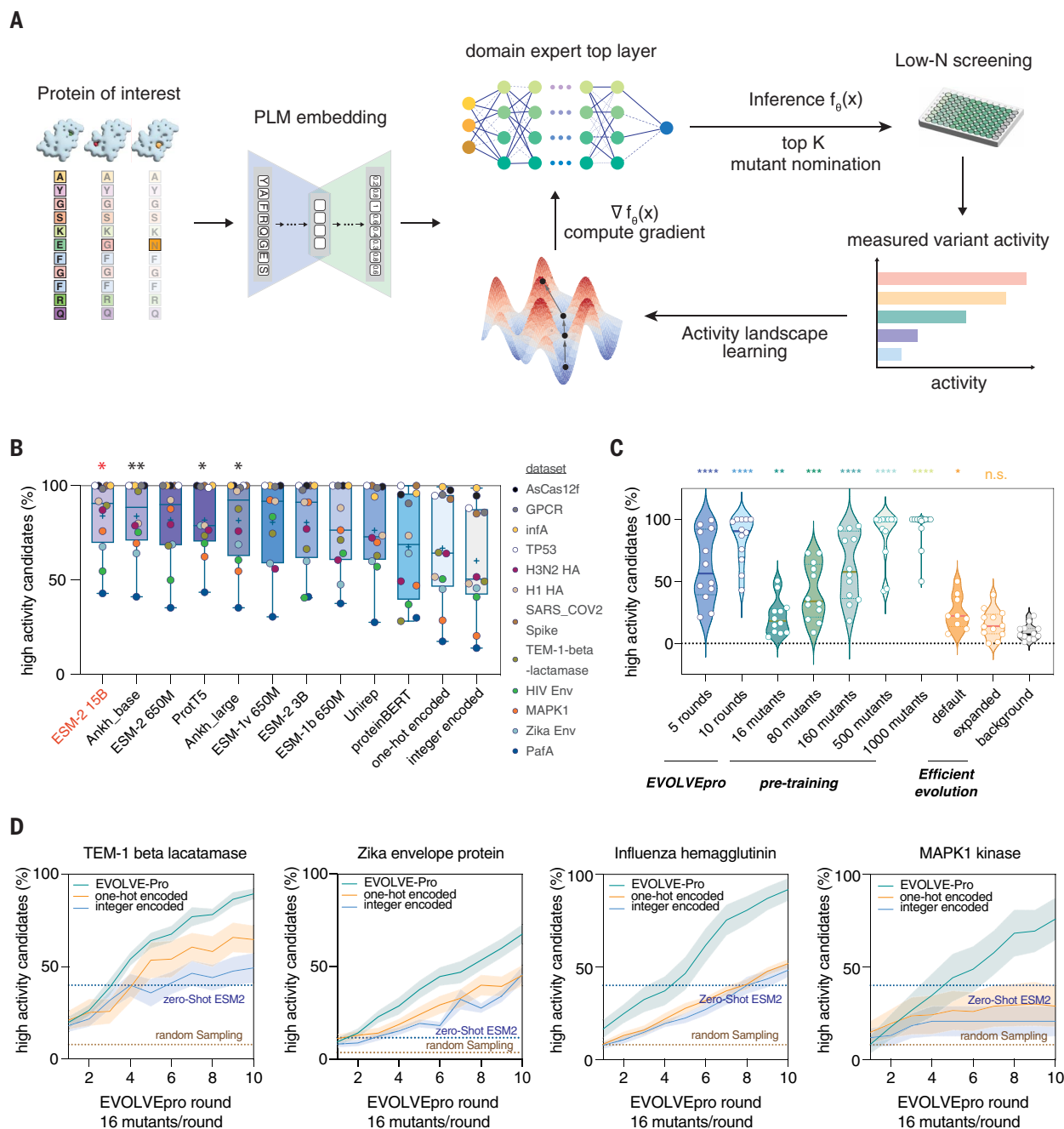


Fig. 1. Developing and benchmarking EVOLVEpro for PLM-guided engineering.

(A) Schematic describing the EVOLVEpro method. Proteins of interest go through iterative rounds of low-N screening. A foundational PLM generates embeddings for all mutants of a protein, and the average embedding by pooling across all residues is used as input for the top-layer model. Each mutant's activity is experimentally determined and used to train a domain expert top-layer model with PLM embedding as input. The top-layer model then nominates the top-N mutants for the next round of testing and the weights are updated iteratively in an active learning format. K, number of mutants to test; $f_{\theta}(x)$, top-layer regression prediction score. (B) Benchmarking of foundational models across a panel of 12 comprehensive DMS datasets. Each point represents a distinct protein

and its DMS data. ESM2-15B has the highest average percent success in high activity variants prediction. (C) Comparison between EVOLVEpro in active learning format, in zero-shot pretraining format, and an existing zero-shot prediction method using PLM (15) across 12 DMS datasets. Each point represents a distinct protein using its DMS data. (D) Performance over 10 rounds of EVOLVEpro with 16 mutants per round, compared to two different non-language model-encoding schemes (one-hot encoding and integer encoding). Model performance is benchmarked on four datasets (31, 36, 40, 42) and compared with zero-shot ESM2 nomination success rate and background random sampling (15). Error bar represents the standard deviation for $n = 10$ random simulations. N.s., not significant; * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$; **** $P < 0.0001$.

actively learns the fitness landscape across multiple rounds of evolution. In each round, the regression model ranks protein sequences according to their predicted activity, selecting

top-ranked sequences for experimental validation. Cycles are performed iteratively to improve defined protein activities until they reach desired levels (Fig. 1A).

We first optimized EVOLVEpro's computational framework in silico by curating 12 DMS datasets (31–43) (table S2 and data S2), allowing the selection of optimal architecture and

parameters by using simulated runs prior to any experimental testing. This simulation revealed ground truth activity data to EVOLVEpro for only the variants nominated by the model. The 12 DMS datasets selected for model benchmarking span diverse activities, including viral spike proteins, RNA-guided nucleases, DNA-binding proteins, RNA-binding proteins, and kinases, maximizing the generalizability of the model architecture, which would serve as the final EVOLVEpro model for experimental applications throughout the rest of the study.

We optimized the EVOLVEpro architecture across five parameters: (i) the strategy for first-round mutant selection, (ii) the top-layer regression model that learns the activity landscape, (iii) the active learning policy for selecting mutants for the next round, (iv) the data processing for experimentally measured activities, and (v) the PLM embedding vector transformation (table S1 and data S1). We first selected ESM-2 as the base PLM because of its large training data and available model size of >200-M proteins and 15B parameters, respectively. Using the ESM-2 15B parameter model, our grid search (see methods) found that the optimal strategy was (i) selecting a random set of first-round variants, (ii) using a random forest regressor discriminatory model to predict protein activities, (iii) using embeddings averaged across all amino acids, and (iv) using a top-N selection strategy in each round of evolution (fig. S1A). This model nominated high frequencies of gain-of-function protein variants in only five rounds (fig. S1, A and B), and both median activity and activity of the nominated top mutant increased rapidly from round to round across all DMS datasets (fig. S1B). We calculated the top mutant's improvement in activity by scaling the activity score of the best mutant in the last round relative to the first round across the 12 DMS datasets. In general, 10 rounds of EVOLVEpro evolution with 16 mutants per round identified top mutants with up to 2.2-fold higher activity than the starting WT sequence (fig. S1B). To understand how the number of variants per round affected performance, we simulated EVOLVEpro evolution with 10 to 100 variants per round. We found that larger rounds increased prediction accuracy without saturation (fig. S1C), indicating that EVOLVEpro can be used for both extremely low-N evolution (<20 mutants per round) for rapid and cheap experimental characterization and medium-N (~100 mutants per round) for quicker and more efficient evolution with fewer rounds.

After optimizing the top-layer model and learning strategies, we surveyed a panel of PLMs. Using the optimal parameters from the grid search, we compared ESM-2 15B with smaller versions of ESM-2 and ESM-1 (44), UniRep (29, 30), ProtT5 (45), ProteinBERT (10), Ankh (9), one-hot encoding, and integer-

encoded protein representations for the ability to identify the highest activity candidates across the 12 datasets. The ESM-2 15B parameter model outperformed other models on all datasets except two and returned the greatest fraction of high-activity mutants, confirming its final selection for the EVOLVEpro latent space model (Fig. 1B and data S3). Across our panel of embedding methods, only four PLMs had a significantly higher prediction accuracy than one-hot encoding, as determined by one-way analysis of variance (ANOVA), showing the critical importance of the base-layer model for EVOLVEpro performance. Given the high dimension of ESM2-15B and the small number of samples seen by the top-layer regression model, we explored whether the full input dimension was needed for our observed model performance. Reducing the input dimension of the protein embedding with PCA, we tested a range of input dimensions into the top-layer regressor. The original full-length embeddings performed best on 9 of the 12 datasets (fig. S1D), with full-length embeddings contributing more to model accuracy in difficult engineering tasks in which there exist fewer high-activity mutants in the population, such as mitogen-activated protein kinase 1 and PafA. By contrast, for easier tasks, such as infA and AsCas12f, all dimensionalities saturated activity.

Because MLDE methods require pretraining a discriminatory model, we compared augmented EVOLVEpro with various amounts of pretraining (Fig. 1C). Active learning drastically reduced the overall number of mutants required: EVOLVEpro with only five rounds of evolution (16 mutants per round) was equivalent in performance to EVOLVEpro pretrained with 160 mutants, whereas 10 rounds of evolution (16 mutants per round) was equivalent to pretraining with 500 mutants. Moreover, EVOLVEpro outperformed zero-shot prediction methods (Fig. 1C) (15). This comparison confirms that the few-shot nature of EVOLVEpro allows for efficient directed evolution with minimal effort and low-N testing per round (Fig. 1C and data S4). To explore whether EVOLVEpro benefited from base models that are better at extrapolation of complex landscapes, we compared the performance of a random forest regressor with a Gaussian process regressor and a k -nearest neighbor-based regressor, finding that the random forest regressor performed best in 10 out of 12 datasets (fig. S1E). This finding agrees with the utility of random forest regressors in the low-N regime in other prediction tasks (46, 47).

Lastly, we analyzed the per-round activity improvement for EVOLVEpro compared to one-hot and integer encoding and zero-shot prediction. With 16 mutations per round, EVOLVEpro found variants with significantly enhanced activity by round five across every dataset (Fig. 1D and fig. S2). Moreover, the one-hot and integer

encoding frameworks often saturated much earlier in the evolution process and never reached the activity levels achieved by EVOLVEpro. Notably, we observed a nonlinear increase in protein activity after round three for some proteins, suggesting more substantial gains in mapping the protein activity landscape as EVOLVEpro evolution proceeds.

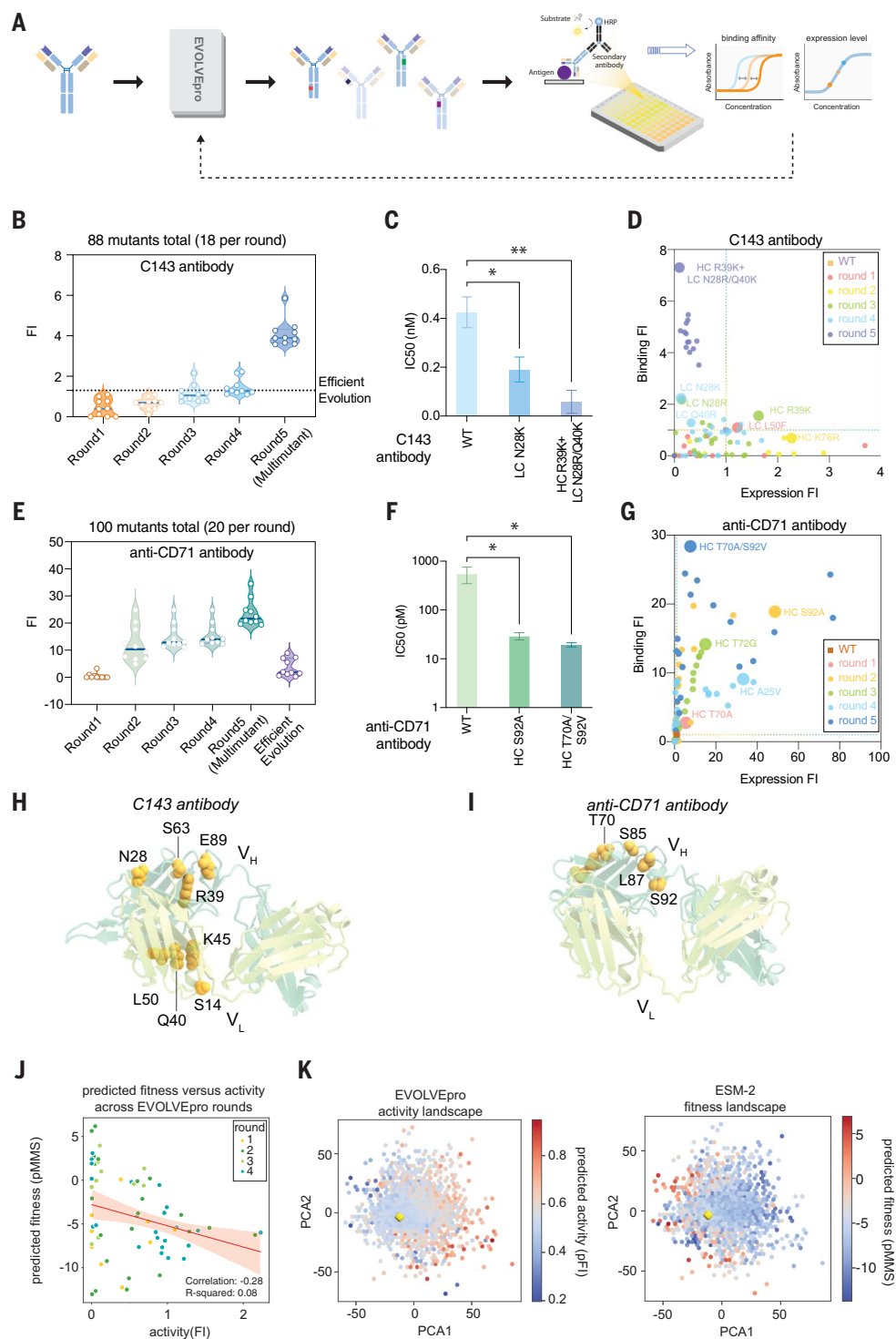
Antibody optimization with EVOLVEpro

We used EVOLVEpro to optimize two therapeutically relevant monoclonal antibodies: C143, an antibody against the SARS-CoV-2 spike protein, and aCD71, an antibody against the human transferrin receptor used for delivery of drugs and small interfering RNA to muscle and cardiac cells in vivo (15, 48, 49). aCD71 has more than 90% sequence homology to Delpacibart, a phase II clinical-stage therapy for Myotonic dystrophy. Both antibodies have low nanomolar affinities against their cognate antigen, presenting a challenge for further improvement by EVOLVEpro. We designed a multiobjective optimization with EVOLVEpro on antibody expression levels and binding affinity to the target antigen. Optimization over multiple features shows that the model can jointly evolve antibody binding and yield, as antibody mutations often affect multiple antibody features, including expression, stability, solubility, half-life, or immunogenicity. Notably, zero-shot developability optimization is difficult with protein language- or structure-based inverse folding models, as the relationship between sequence and developability or other nonbinding features is not directly captured by evolutionary sequence or structure data (15, 20). In our multiobjective directed evolution scheme, we weighed the binding affinity at four times the expression level (i.e., developability score) to prioritize variants that bind with higher affinity (Fig. 2A).

For C143 evolution, we quantified binding affinity with an enzyme-linked immunosorbent assay (ELISA) against the SP6 stabilizing variants of the SARS-CoV-2 spike (Wuhan strain) protein (50). We saw improved binding after three rounds of EVOLVEpro evolution, surpassing previous zero-shot approaches (15) (Fig. 2B and fig. S3A). At round four, we found significant improvement with a light chain mutant (N28K), with a median inhibitory concentration (IC_{50}) of 0.19 nM (Fig. 2C). Using our four rounds of single mutations, we had EVOLVEpro design multimutant combinations for a fifth round. The best multimutant (light chain Asn²⁸Arg/Gln⁴⁰Lys with heavy chain Arg³⁹Lys) bound to the SP6 spike antigen with an IC_{50} of 60 pM (Fig. 2C), likely owing to the synergistic interaction between Asn²⁸Arg on the light chain and Arg³⁹Lys on the heavy chain. We stopped at one round of multimutant evolution owing to the substantial improvements we observed, but in practice, multiple rounds of multimutant

Fig. 2. Evolution of two monoclonal antibodies with EVOLVEpro.

(A) Schematic of the evolution strategy with EVOLVEpro for engineering two monoclonal antibodies across two parameters (binding affinity and antibody expression). HRP, Horseradish peroxidase. **(B)** Engineering of the C143 antibody over five rounds of EVOLVEpro. Data shows cumulative top 10 mutants' fold improvement over WT binding affinity to the target antigen across five evolution rounds. **(C)** IC₅₀ value estimated from ELISA binding data for the WT C143 antibody, the best single mutant (LC Asn²⁸Lys) and the best multimutant (LC Asn²⁸Lys/Gln⁴⁰Lys + HC Arg³⁹Lys). Error bars represent standard error of mean with $n = 3$ technical replicates. A one-way ANOVA was run between the three groups (* $P < 0.05$; ** $P < 0.01$). **(D)** Scatter plot showing each individual mutant's expression fold improvement (FI) versus binding affinity fold improvement for the C143 antibody. The best mutant in each round is highlighted with a larger circle. **(E)** Engineering of the aCD71 over five rounds of EVOLVEpro. Data shows cumulative top 10 mutants' fold improvement over WT binding affinity to the target antigen across five evolution rounds. **(F)** IC₅₀ value estimated from ELISA binding data for the WT anti-CD71 antibody, the best single mutant (Ser⁹²Ala), and the best multimutant (Thr⁷⁰Ala_Ser⁹²Val). Y axis is shown on log₁₀ scale. Error bars represent standard error of mean with $n = 3$ technical replicates. A one-way ANOVA was run between the three groups (* $P < 0.05$; ** $P < 0.01$). **(G)** Scatter plot showing each individual mutant's expression fold improvement versus binding affinity fold improvement for the aCD71 antibody. The best mutant in each round is highlighted with a larger circle. **(H and I)** Mapping of the top mutations on the predicted structure of C143 (H) and anti-CD71 (I), respectively. **(J)** Scatter plot comparing the predicted naive ESM-2 C143 protein fitness (pMMS) and scaled tested activity of nominated mutants across evolution. Scatter points are colored by rounds in evolution. The correlation and linear regression line are shown in red, and the coefficient of determination (R^2) of the correlation is reported. **(K)** Comparison of the C143 embedding latent space with either predicted naive ESM-2 protein fitness landscape or EVOLVEpro protein activity landscape. The yellow rhombus denotes WT sequence. Single-letter abbreviations for the amino acid residues referenced throughout the figures are as follows: N, Asn; K, Lys; R, Arg; Q, Gln; L, Leu; F, Phe; T, Thr; A, Ala; S, Ser; V, Val; G, Gly; E, Gl; I, Ile; Y, Tyr; M, Met; D, Asp; P, Pro; H, His.



testing are likely needed to reach convergence on desired properties. We found that many improved binders compromised yields (Fig. 2D), a tradeoff due to the bias toward binding during the multiobjective design. Despite this tradeoff, a subset of C143 mutants, such as Arg³⁹Lys,

had both an increase in affinity and protein expression, showing that developability can be co-optimized alongside binding affinity.

We explored the likelihood of the top EVOLVEpro-nominated mutations relative to training data and known antibody variants observed in nature.

We analyzed the top 10 mutations for both occurrence in hotspot regions and deviation from the germline sequence. We found that none of the top 10 mutations were mutated back to the germline unmutated common ancestor (UCA) sequence. The UCA sequence at light chain Asn²⁸

is a serine. However, the affinity-enhancing mutation recommended by EVOLVEpro is either a lysine or arginine, both with likelihoods of less than 0.05 when compared with the Uniprot training input, reinforcing the notion that EVOLVEpro mutations are rare and newly discovered. Moreover, this observation highlights the utility of the top-layer regression model to explore rare mutations not seen in the training input of PLM by promoting exploration into unknown regions of the protein fitness landscape. Furthermore, we found the top single mutant, Asn²⁸Arg (light chain), happens in the complementarity determining region (CDR), but most of the top affinity-enhancing mutations (7 out of the top 10 mutations) are located in the framework region. This highlights the de novo exploration of EVOLVEpro on the entire variable region of the antibody to find affinity-enhancing mutations that might not seem likely or intuitive in the framework region. To further understand EVOLVEpro's mutational trajectory, we represented the model's attention to particular residues as a cumulative frequency and found that residues such as Lys³³, Arg³⁹, and Asp⁵⁸ on the heavy chain and Ser¹⁴ and Asn²⁸ on the light chain are repeatedly explored (fig. S3B).

For using EVOLVEpro to *in silico* evolve an anti-CD71 antibody, we measure the target binding affinity using ELISA against the human TfR protein and measured antibody expression with an anti-IgG. We saw improvement in binding after just rounds of evolution with EVOLVEpro (Fig. 2E and fig. S4A). We acquired 10 mutants using the efficient evolution algorithm to benchmark against EVOLVEpro and found that EVOLVEpro nominated mutants 35-fold better than WT whereas efficient evolution's best mutant is only eight-fold better (15). At round 4, we found the best single mutant heavy chain S92A to bind to the antigen with an IC₅₀ of 29 pM, significantly higher than that of the WT at 551 pM (Fig. 2F). We also asked the model to rank multi-mutants based on the single mutant data from the first four rounds and performed one round of multi-mutant testing. We improved binding and expression in the multimutant round five with heavy chain Thr⁷⁰Ala/Ser⁹²Val mutant. The multimutant binds to the hTfR protein with an IC₅₀ of 19 pM (Fig. 2F). Notably, most of the mutants nominated after round one showed a marked increase in expression profile and binding affinity, showing that EVOLVEpro simultaneously engineered the developability and binding (Fig. 2G). This finding contrasts with the results from the C143 antibody, implying the Pareto frontier of binding and expression likely differs between the two antibodies, with anti-CD71's WT sequence easier to engineer across multiple properties than C143's WT sequence. Future work examining this trade-off between multiple properties for additional anti-

bodies or proteins will enable EVOLVEpro to better traverse Pareto frontiers more efficiently.

Upon analyzing the novelty of the aCD71 antibody mutations, we found that only one of the top 10 mutations is mutated back to the germline UCA sequence at position Val⁷³. The UCA sequence at the site of the best mutation in heavy chain Ser⁹² is a threonine. However, the affinity-enhancing mutation recommended by EVOLVEpro is either an alanine or valine. Ser⁹²Val mutation has a mutation likelihood of less than 0.05 when compared with the Uniprot training input, highlighting its rarity. This shows EVOLVEpro's ability to insightfully choose new mutations not seen in the training input of the PLM base layer. Furthermore, we found that all top 10 affinity-enhancing mutations are located in the framework region rather than in the CDR that is commonly thought to determine binding affinity. Lastly, to understand EVOLVEpro's mutational trajectory on anti-CD71, we represented the model's attention to particular residues as the cumulative frequency of individual residues being explored by the model and found that multiple residues are repeatedly explored by the model including Thr⁷⁰ and Ser⁹² on the heavy chain and Gln³⁸ on the light chain (fig. S4B).

We used AlphaFold 3 to model the structure of the anti-CD71 and C143 antibodies (Fig. 2, H and I, and data S7). We found two major clusters of exploration by EVOLVEpro on C143 antibody in the framework region with light chain mutations Ser¹⁴, Gln⁴⁰, Leu⁵⁰, and Lys⁴⁵ co-located and Arg³⁹, Ser⁶³, and Glu⁸⁹ in close proximity on the heavy chain. These mutations likely alter binding through structural changes in the variable region. Additionally, it was a CDR mutation, Asn²⁸, on the light chain located in the CDR-L1 region that likely directly alters the interaction between the C143 antibody and the antigen, which is not possible to model with AlphaFold 3 because of a low confidence score of the complex (Fig. 2H). For the anti-CD71 antibody, we found all the best mutations clustered around one region in the heavy chain domain. Because they are all in the framework region, they likely alter the binding affinity indirectly, a hypothesis supported by the increase in expression relative to the WT sequence (Fig. 2I).

Lastly, we analyzed each mutant's observed activity versus the PLM-predicted fitness landscape. We calculated the mutant fitness as a predicted masked marginal score (pMMS) within the ESM2 embeddings and found that the activities of EVOLVEpro variants did not correlate with predicted ESM2 fitness (Fig. 2J and fig. S4C). To extrapolate this finding across the entire C143 and anti-CD71 mutational landscape, we projected the base-layer PLM fitness score and top-layer random forest predicted fold improvement (pFI) in the latent space for every possible single mutant variant, generating EVOLVEpro-determined protein activity land-

scape (figs. S3C and S4E). There was relatively little overlap between the two distributions, with a negative correlation of -0.16 for C143 antibody and 0.01 for anti-CD71 antibody between predicted fitness and predicted activity, further highlighting ESM2's lack of understanding of protein activity (figs. S3C and S4E). We projected the individual mutants onto the PCA space of the ESM2 embedding and found two opposing directions between higher fitness and higher function (Fig. 2K and fig. S4D). Analyzing the evolution trajectory from round 1 to the final round by calculating the geometric midpoint of each round revealed that the top-layer model directed the evolutionary process toward the higher side of PCA1 for C143 antibody and the higher side of PCA2 for anti-CD71, which we observed to correlate with higher protein function (figs. S3D and S4F).

Evolution of a miniature RNA-guided CRISPR nuclease with EVOLVEpro

Programmable RNA-guided nucleases have diverse applications in basic biology, therapeutics, and diagnostics. However, commonly used nucleases, such as the Cas9 from *Streptococcus pyogenes* (SpCas9), are too large to effectively be packaged in common adeno-associated viral (AAV) vectors and more compact high-efficiency nucleases, such as the Cas9 from *Staphylococcus aureus* (SaCas9), still preclude the use of larger regulatory elements or protein fusions. Miniature Cas12f nucleases have compact sizes (<700 residues) but suffer from reduced efficiencies, requiring engineering for genome editing applications (51). Previous Cas12f engineering efforts relied on DMS or rationally designed mutations to increase the *in vitro* cleavage activity (33, 52–55), requiring extensive screening to find the optimal variant. We tested whether EVOLVEpro could rapidly develop highly active Cas12f variants to accelerate miniature nuclease engineering.

We selected the Cas12f from *Pseudomonas aeruginosa* (PsaCas12f) for evolution with set indel formation at the endogenous *RNF2* locus target site as the optimization metric (Fig. 3A). After four rounds of evolution of 12 single mutants per round, EVOLVEpro yielded point mutants of PsaCas12f with up to 4.9-fold improvement in indel formation. This top variant, PsaCas12f^{Lys333Val}, had >40% indel efficiency at the *RNF2* site (Fig. 3B and fig. S5A). To identify synergies between EVOLVEpro-nominated mutants, we combined the top-performing variants from previous rounds in a fifth round. We evaluated a set of these multimutants and found that PsaCas12f^{Ile178Ala/Lys333Val/Lys454Pro} generated ~50% indel activity at the *RNF2* locus (fig. S5A). Given its performance, we refer to the PsaCas12f^{Ile178Ala/Lys333Val/Lys454Pro} variant as EVOLVEpro PsaCas12f (epPsaCas12f).

To generalize epPsaCas12f's improved activity, we evaluated the enzyme at 10 different

Fig. 3. Evolution of highly active miniature CRISPR nucleases with EVOLVEpro.

(A) Schematic of the evolution strategy with EVOLVEpro for engineering a miniature Cas12f.

(B) Engineering of PsaCas12f over five rounds of EVOLVEpro, with the last round being multi-mutant.

Data shows cumulative top 10 mutants from current and preceding rounds, as measured by fold improvement of indel activity at the endogenous RNF2 genomic locus. **(C)** Indel activities of WT PsaCas12f, epPsaCas12f, and a panel of published Cas12a and Cas12f nucleases on 10 different genomic targets across five genes (two guides per gene).

The fold change on top of each guide denotes the relative fold increase of epPsaCas12f compared with the average of the other published Cas12a and Cas12f nucleases. A one-way ANOVA was performed for each guide sequence shown (**** $P < 0.0001$). Error bars represent standard error of mean with $n = 3$ biological replicates.

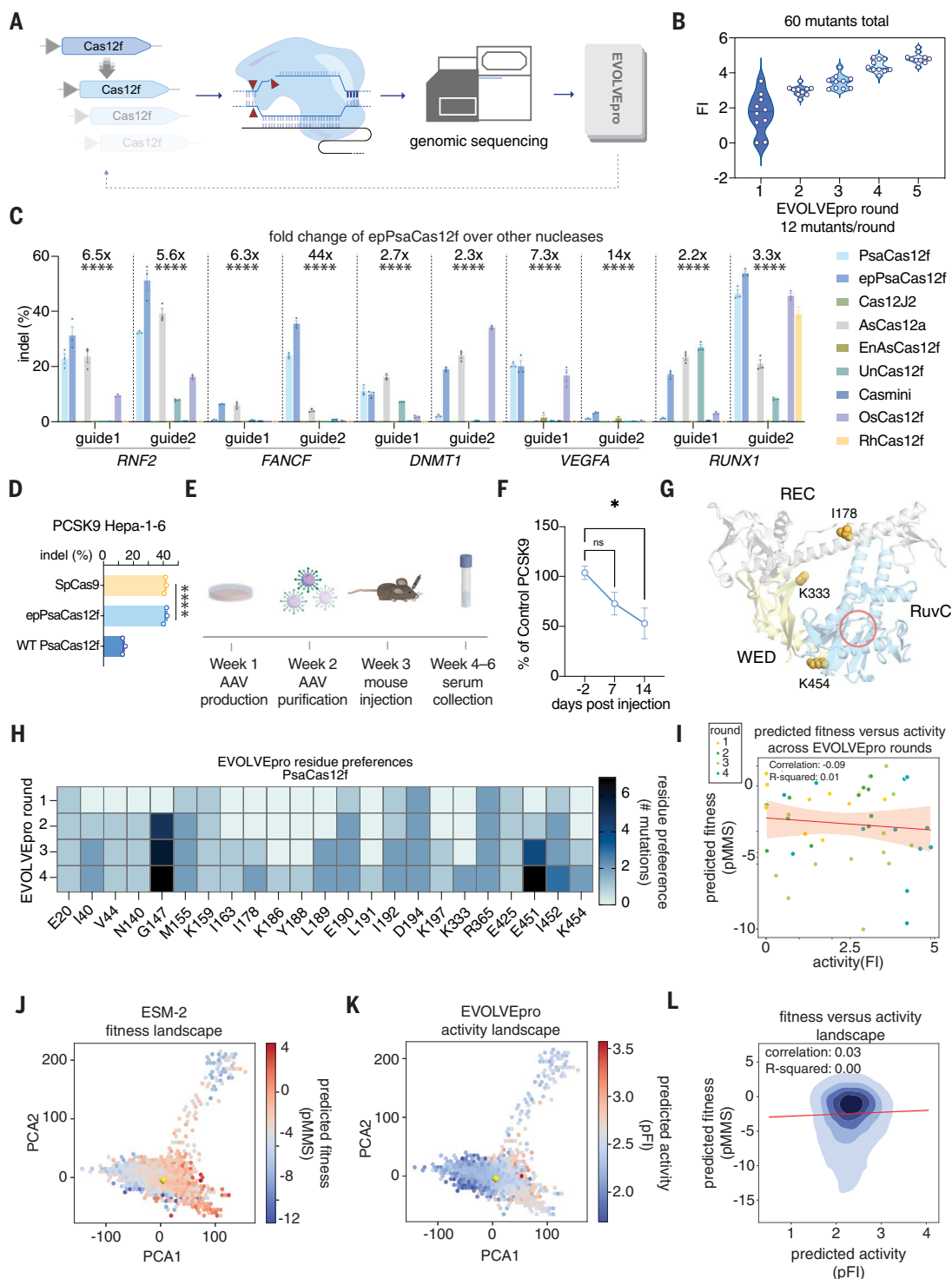
(D) Next-generation sequencing quantified indel formation at murine PCSK9 genomic loci by epPsaCas12f, WT PsaCas12f, and SpCas9. A one-way ANOVA was performed for each guide sequence shown (**** $P < 0.0001$). Error bars represent standard error of mean with $n = 3$ biological replicates.

(E) Schematic of the in vivo validation assay for EnPsaCas12f editing at the murine PCSK9 locus for PCSK9 reduction.

(F) Serum PCSK9 levels at three different time points from -2 days of injection to +14 days. The percentage of control PCSK9 was calculated by normalizing to the control group with phosphate-buffered saline (PBS) injected. A two-sided Student's t test was run on each time point relative to -2 days' baseline PCSK9 level (ns, not significant; * $P < 0.05$). Error bars represent standard error of mean with $n = 3$ biological replicates.

(G) Mapping of the top mutations on the AlphaFold 3 model of PsaCas12f. The RuvC active site is indicated by a red circle. **(H)** Heatmap showing most common PsaCas12f mutations explored by EVOLVEpro over rounds of evolution. Any position explored more than once is shown on a cumulative basis across rounds.

(I) Scatter plot comparing the predicted naive ESM-2 protein fitness (pMMS) and



targets across five endogenous genomic loci, comparing with WT *PsaCas12f* and seven previously characterized Cas12 effectors, *AsCas12a*, *Cas12Φ*, *UnCas12f1*, *enAsCas12f*, *OsCas12f*, *RhCas12f*, and *CasMINI* (33, 53, 55–58). We observed consistently higher *epPsaCas12f* activity compared with that of WT *PsaCas12f* on 9 of 10 tested targets (Fig. 3C). Moreover, *epPsaCas12f* edited the 10 targets with a $23.3 \pm 16.7\%$ average indel rate, surpassing all tested miniature Cas12f effectors and *AsCas12a* with 2.2- to 44-fold improvement. Notably, *epPsaCas12f* generated an average deletion of 5 base pairs (bp) across the 10 tested targets (fig. S5B). Together, these data demonstrate that *epPsaCas12f* is a highly active, compact effector for mammalian genome editing that outperforms other small effectors.

We applied *epPsaCas12f* for in vivo genome editing applications, using its compact size for single-vector viral delivery in vivo. We designed guides targeting a sequence 5' of exon 3 in the mouse *PCSK9* gene (Fig. 3D). The *PCSK9* protein regulates blood low-density lipoprotein (LDL) by binding to LDL receptors, making it a valuable therapeutic target (59). We first tested the efficacy of *epPsaCas12f* in a murine hepatocyte cell line (Hepa 1-6) by cotransfecting murine codon-optimized *epPsaCas12f* and single guide RNA (sgRNA) targeting sequences 5' of exon 3 in the *PCSK9* gene. Analyses of *epPsaCas12f*, WT *PsaCas12f*, and *SpCas9* revealed that *epPsaCas12f* robustly edited *PCSK9* in Hepa1-6 cells with ~40% indel formation (comparable levels to *SpCas9* and threefold higher than the WT *PsaCas12f*) (Fig. 3D).

After validation of *epPsaCas12f* in Hepa1-6 cells, we packaged both *epPsaCas12f* and its sgRNA targeting *PCSK9* in a single AAV2/8 vector (Fig. 3E). AAV-*epPsaCas12f* was administered at a titer of 1.5×10^{12} viral genome copies per mouse via retro-orbital injection into 3-month-old C57BL/6J mice. We tracked blood *PCSK9* levels for 14 days after the injection of AAV and found a significant decrease to around 50% of the original levels after 14 days (Fig. 3F and fig. S5C). We then harvested the liver at day 15, isolated the genomic DNA, and performed next-generation sequencing to survey for indel formation at the *PCSK9* target site (Fig. 5, D and E). We found around 7% on-target indel formation in the AAV-*epPsaCas12f*-injected mice (fig. S5D), demonstrating that *epPsaCas12f* can be used for single-vector AAV-mediated genome editing. To survey off-targets, we used Cas-OFFinder to predict the top four off-target cleavage sites generated by *epPsaCas12f* and analyzed the guide-dependent off-target cleavage in the liver (60). We only found detectable editing at one of the four sites with a maximum level of 0.27% indels, confirming minimal off-target cleavage triggered by *epPsaCas12f* (fig. S5F).

To understand the mechanisms of the beneficial mutations nominated by the EVOLVEpro,

we used AlphaFold 3 to predict the structure of *PsaCas12f* (Fig. 3G and data S7). The predicted structure provides insight into how the PLM-nominated mutations, including Ile¹⁷⁸Ala/Lys³³³Val/Lys⁴⁵⁴Pro, contribute to enhancing the DNA cleavage activity (Fig. 3G). The Lys³³³Val mutation is located in the WED domain, suggesting that it could increase the binding to its RNA guide. The Ile¹⁷⁸Ala mutation is located in the middle of the long α helix in the REC domain and forms a hydrophobic core with Ile²⁴⁵ and Leu²⁴⁸ in the adjacent α helix. Given that alanine is a helix-forming residue, the Ile¹⁷⁸Ala mutation may stabilize the α helix in the REC domain and thus augment the cleavage activity. The Lys⁴⁵⁴Pro mutation is located at the C terminus of an α helix in the RuvC domain and forms hydrophobic interactions with Ala⁵⁰⁹ and Val⁵¹¹ in the adjacent α helix, suggesting that it also stabilizes the protein conformation.

We then looked at the model's attention to particular residues in the protein by calculating the cumulative frequency of individual residues explored by the model. Multiple residues were repeatedly nominated by the model, including Gly¹⁴⁷ and Glu⁴⁵¹ (Fig. 3H), showing that the model honed attention to specific amino acids. We calculated the pMMS for each nominated mutant to understand the relationship between the base-layer PLM's fitness prediction and the actual measured protein activity (Fig. 3I). We found a weak negative correlation between fitness and activity in *PsaCas12f*'s local context. We then further projected the base-layer PLM's fitness score and the top-layer random forest regressor's activity score in the EMS2 latent space to understand EVOLVEpro's global mutational trajectory (Fig. 3, J to L, and fig. S5G). We found a weak positive correlation of 0.03 between fitness and activity, further denoting the necessity of a top-layer discrimination model to properly distinguish between high fitness and high activity (Fig. 3I).

Engineering improved prime editors with EVOLVEpro

Many molecular tools, such as next-generation genome editing proteins, function as multiple enzymes acting in concert. Prime editing (PE), which uses an RNA-templated reverse transcriptase to programmably install diverse genome edits, is the fusion of a *SpCas9* nicking mutant (nCas9) with an engineered Moloney Murine Leukemia Virus Reverse Transcriptase (M-MLV RT) (Asp²⁰⁰Asn, Leu⁶⁰³Trp, Thr³⁰⁶Lys, Trp³¹³Phe, Thr³³⁰Pro) (termed PE2). We reasoned that EVOLVEpro could improve upon these rational mutations, as optimizations were discovered on M-MLV RT by directed evolution approaches (61). Because PE-based insertion has difficulty installing longer (>40-nt) edits, we focused on editing outcomes with longer (46-bp) insertions, which have particular utility for programmable

gene insertion methods, such as programmable addition via site-specific targeting elements (PASTE) (62). We set up the evolution policy with a previously described twinPE approach, where two overlapping prime editing guide RNAs are used in combination to install a 46-bp attB site in the *NOLC1* loci in murine hepatocyte cell line (Hepa1-6). Editing was quantified at *NOLC1* loci using amplicon sequencing and next-generation sequencing read-out, and the top-layer EVOLVEpro model was trained to predict the insertion efficiency.

Over successive rounds of optimization, we found that EVOLVEpro progressively learned the activity landscape of the RT of PE2, yielding improved variants after the initial random selection round and substantially improving upon PE2-based editing by round four (Fig. 4A and fig. S6A). To check for bias toward this single locus in the genome that could have developed during the evolution campaign, we tested the top four performing variants (Ala⁶⁶⁰Ser, Leu⁶⁷⁰Cys, Leu⁶⁷⁰Lys, and Leu⁶⁷¹Arg) at three additional genomic loci (human *AAVS1*, human *ALB*, and mouse Factor IX) in two cell lines. At all four sites tested, Ala⁶⁶⁰Ser had statistically significant improvements (Fig. 4B). These results point to the general protein activity improvement by EVOLVEpro, delivering an additional set of RT mutations specifically for larger edits.

Projecting the top mutations onto the AlphaFold 3–predicted structure of the RT reveals that most of them are clustered in the C-terminal RNA polymerase H (RNase H) domain (Fig. 4C), which is a surprising result, as most PE evolution focuses on RT mutagenesis. We speculate that these mutations could alter the cleavage of the template DNA in the RNA-DNA heteroduplex by the RNase H domain (63), facilitating the completion of the prime editing reaction, a route that has not been explored by traditional engineering of prime editors. Alternatively, they could inactivate the RNase H domain, as truncations of this domain have been shown to slightly improve prime editing activity (64–66). We then further analyzed EVOLVEpro's residue site preference during evolution and observed attention to residues such as Leu⁶⁷⁰, Leu⁶⁷¹, and Ala⁶⁶⁰, suggesting that it was learning that these positions could be quite beneficial for improving activity (Fig. 4D). Analysis of predicted fitness (pMMS) scored by the bottom layer PLM again showed a divergence between fitness and activity for the prime editor (Fig. 4E and fig. S6B), as we found almost no convergence between the two distributions with a negative correlation of 0.08 (Fig. 4E).

We tried to understand the global mutational trajectory by projecting the activity landscape learned by the random forest regressor and base layer ESM2's protein fitness landscape onto the first two PCAs of the embedding (fig. S6, C and D). This analysis points again to the

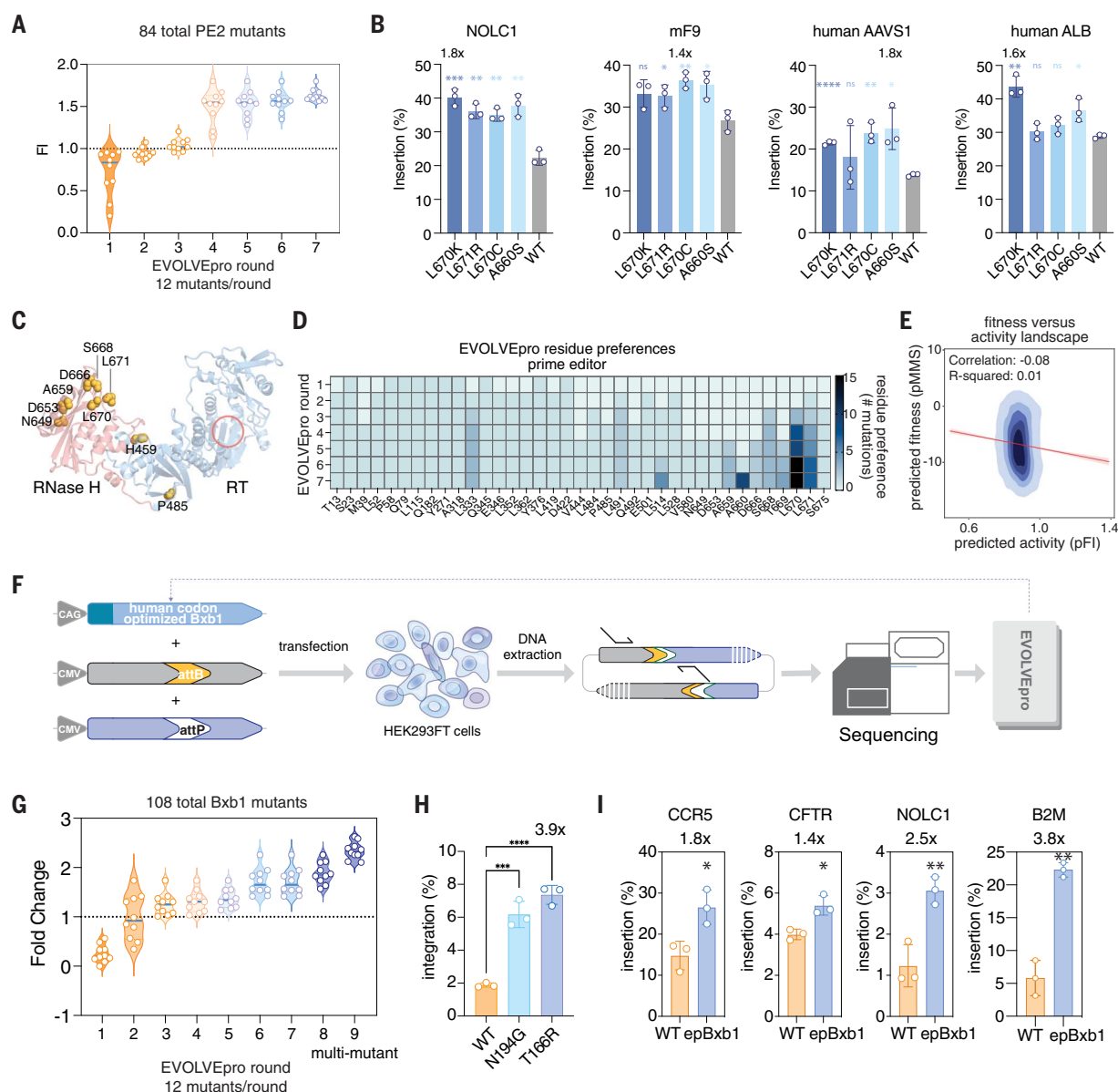


Fig. 4. Evolution of prime editor with EVOLVEpro. (A) Engineering of the prime editor PE2 with twinPE guides over seven rounds of EVOLVEpro. Data shows cumulative top 10 mutants from current and preceding rounds, as measured by fold improvement of prime editing activity to install a 46-bp AttB site at the murine NOLC1 genomic locus. (B) Validation of four evolved prime editors in the installation of attB sites at four different endogenous sites in either mouse or human genomes. A two-sided unpaired *t* test was run between WT and each evolved prime editor (ns, not significant; **P* < 0.05; ***P* < 0.01; ****P* < 0.001; *****P* < 0.0001). Fold change over WT PE2 is shown for the best mutant on each genomic locus. Error bars represent standard deviation with *n* = 3 biological replicates. (C) Mapping of the top mutations on the AlphaFold 3 model of M-MLV RT. The RT active site is indicated by a red circle. (D) Heatmap showing most common PE2 mutations explored by EVOLVEpro over rounds of evolution. Any position explored more than once is shown on a cumulative basis across rounds. (E) Scatter plot comparing the predicted naive ESM-2 protein fitness

(pMMS) and scaled tested activity of nominated mutants across evolution; scatter points are colored by rounds in evolution. (F) Schematic of the evolution strategy for evolving the Bxb1 serine integrase from the mycobacteriophage. (G) Engineering of the Bxb1 integrase over eight rounds of EVOLVEpro. Data shows cumulative top 10 mutants from current and preceding rounds, as measured by fold improvement of plasmid integration over WT. (H) Performance of top Bxb1 mutants for plasmid recombination with low Bxb1 expression in Hela cell. A two-sided Student's *t* test was run between WT and each evolved Bxb1 integrase (****P* < 0.001; *****P* < 0.0001). Fold change over WT Bxb1 is shown for the best mutant. Error bars represent standard deviation with *n* = 3 biological replicates. (I) Validation of epBxb1 with PASTE at four genomic sites across human and mice genomes. A two-sided Student's *t* test was run between WT and each evolved Bxb1 integrase (**P* < 0.05; ***P* < 0.01). Fold change over WT Bxb1 integrase is shown for each genomic locus. Error bars represent standard deviation with *n* = 3 biological replicates.

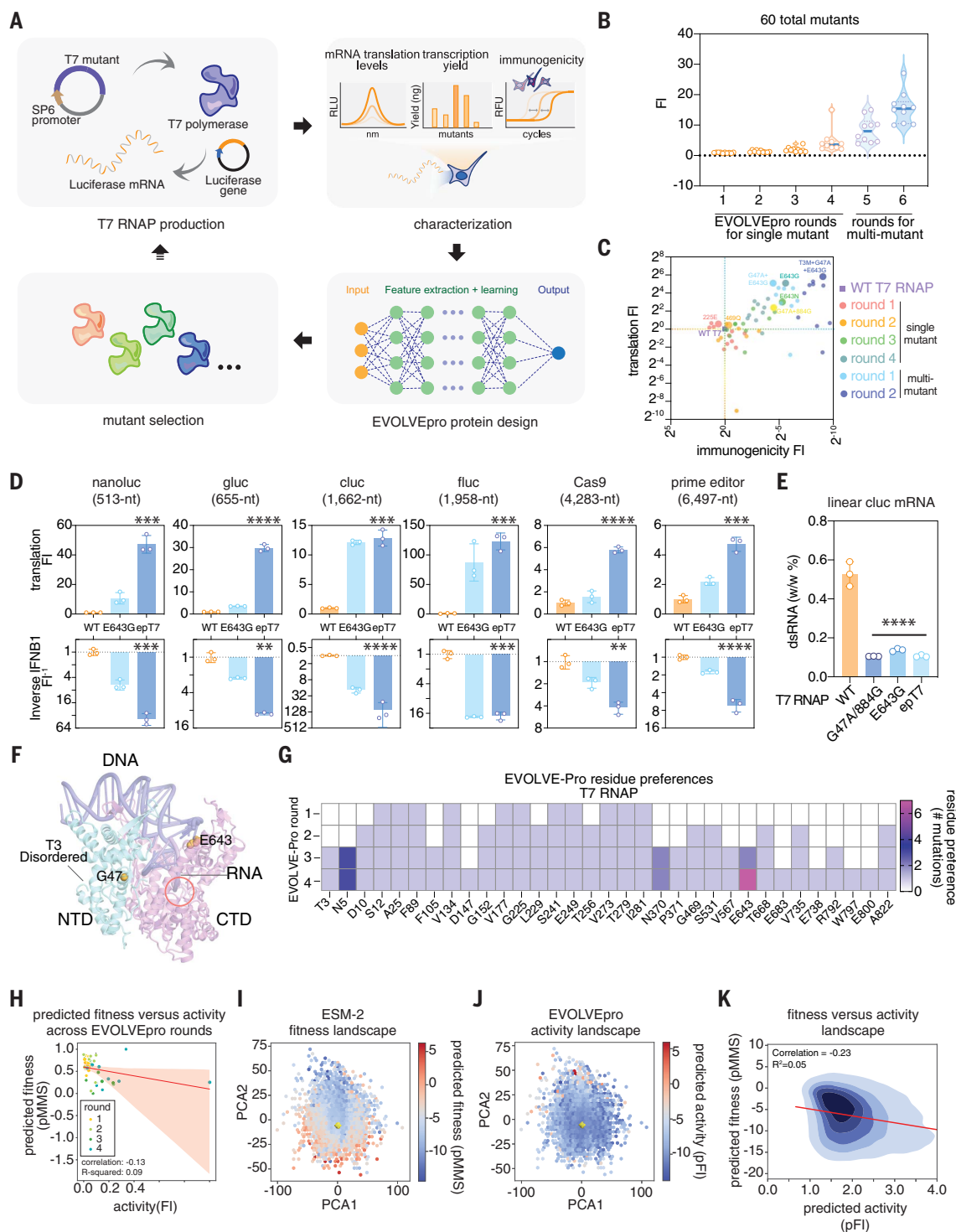


Fig. 5. Engineering RNAPs for high yield and low immunogenicity mRNA production. (A) Schematic of the strategy for high-throughput T7 RNAPs mutant testing and evolution policy setup for evolving a high fidelity T7 RNAP. RLU, relative light units; RFU, relative fluorescence units. (B) Engineering of T7 RNAP over six rounds of EVOLVEpro. Data shows the top 10 mutants from current and preceding rounds, as measured by fold improvement of transcription fidelity over WT. (C) Performance of T7 mutants from six EVOLVEpro rounds and previously engineered Gly⁴⁷Ala/⁸⁸⁴insGly SOTA T7 RNAP in *Cypridina* luciferase (Cluc) mRNA translation and immunogenicity in BJ fibroblast cells. (D) Validation of epT7 for production of six mRNA sequences ranging from 513 nt

to 649 6nt. Purified WT or mutant RNAP is used to produce these sequences, and they were transfected into BJ fibroblast cells for either protein translation readout or targeted IFN- β 1 gene expression analysis using quantitative polymerase chain reaction 24 hours after transfection. A two-sided Student's *t* test was run between WT and each evolved T7 RNAP (***P* < 0.01; ****P* < 0.001; *****P* < 0.0001). Error bars represent standard deviation with *n* = 3 biological replicates. Gluc, *Gaussia* luciferase. (E) dsRNA ELISA was used to analyze the amount of dsRNA during transcription of a 1662-nt Cluc mRNA. 500 ng of posttranscription product was used as input for the dsRNA ELISA. A two-sided Student's *t* test was run between WT and each evolved T7 RNAP (*****P* < 0.0001).

Error bars represent standard deviation with $n = 3$ biological replicates.

(F) Mapping of the top mutations on the T7 RNAP structure (PDB ID: 3E2E). The active site is indicated by a red circle. **(G)** Heatmap showing most common T7 RNAP mutations explored by EVOLVEpro over rounds of evolution. Any position explored more than once is shown on a cumulative basis across rounds.

(H) Scatter plot comparing the predicted ESM-2 protein fitness score versus experimentally measured T7 RNAP transcription fidelity scaled score across

evolution rounds. The correlation and linear regression line are shown in the plot. **(I and J)** Comparison of the T7 RNAP latent space with either predicted

ESM-2 protein fitness (MMS) or EVOLVEpro protein activity fold improvement. Yellow rhombus denotes WT sequence. **(K)** A kernel density estimate of protein

fitness as predicted by ESM-2 versus protein activity as predicted by EVOLVEpro. The correlation and linear regression line are shown in red, and the R^2 of correlation is reported.

divergence between the mutational landscape of a protein's activity and the commonly used fitness landscape learned during a foundational model's training on all protein sequences. When we charted the evolution trajectory, we also found a clear movement to the lower side of PCA2 protein embedding dimension, showing that the top layer is pushing the sequences to that region in the latent space (fig. S6E).

Bxb1 integrase evolution with EVOLVEpro

Large serine recombinases (LSRs) are enzymes that facilitate precise DNA rearrangements, making them crucial tools for genome editing. Their ability to recognize specific DNA sequences and catalyze targeted recombination events allows for efficient and accurate modifications of genetic material, which is essential for advanced gene therapy, synthetic biology, and genetic research. We recently developed a gene insertion technology, PASTE, that leverages LSRs, specifically the Bxb1 integrase, for programmable gene insertion in eukaryotic cells (62). A limitation of Bxb1 integrase, however, is its activity saturates in the 20 to 60% range in cells, limiting the overall integration efficiency that can be achieved. We sought to therefore evolve Bxb1 using EVOLVEpro to improve its activity and demonstrate improved gene integration applications with PASTE in cells.

To evolve Bxb1, we designed a simple integration assay in human embryonic kidney (HEK) 293FT cells that involved the insertion of an AttP-containing DNA plasmid into an AttB target-containing plasmid (Fig. 4F). Integration can be measured by next-generation sequencing, and the evolution policy is designed to optimize this insertion efficiency. We started evolution with a round of 11 random Bxb1 point mutation variants and then over nine rounds observed progressively increasing activity resulting in mutants with more than 2.6-fold higher activity than the WT (Fig. 4G and fig. S7A). As Bxb1 is already fairly active, this fold improvement is expected as we reach near-saturating levels of insertion. To validate the top hits from the evolution campaign, we performed a Bxb1 plasmid titration experiment in a separate cell line (Hela cells) and observed up to fourfold improvement in recombination efficiency under low Bxb1 expression (Fig. 4H). We further validated the top hits by preinstalling attB sites into the genome of HEK293FT cells using lentivirus and then surveyed for

integration efficiency of cargo in the genome. We found up to fourfold improvement by EVOLVEpro's mutants compared to the WT (fig. S7B). We termed Thr¹⁶⁶Arg variant as EVOLVEpro-enhanced Bxb1 (epBxb1).

To test whether the epBxb1 variant's improved activity can improve the programmable insertion of cargo DNA into the chromosome, we tested this variant in the context of PASTE and compared it against the WT Bxb1 across five different genomic loci. We found up to approximately fourfold improvement in the final large cargo insertion rate into the genome, which highlights the generalizable gain in activity (Fig. 4I and fig. S7C).

An AlphaFold 3-predicted model of Bxb1 bound to attachment site DNA indicates that the top beneficial EVOLVEpro mutations clustered in the Bxb1 DNA binding domains, likely increasing the affinity to its DNA targets (fig. S8A). Of these residues, Val²⁹²Ser could directly interact with the phosphate backbone of the target DNA based on its positioning relative to the attachment site, whereas the others likely modulate DNA binding through indirect interactions. Analysis of the residue exploration by the model revealed that multiple positions, including Phe⁴³⁹, Val³⁷⁵, and Leu²⁷⁵, are visited up to 15 times; the DNA-interacting residue Val²⁹² was also visited multiple times. This highlights EVOLVEpro's ability to recognize the functional importance of certain regions in the protein, much like structure-guided engineering approaches (fig. S8B).

We then calculated the relationship between the fitness (pMMS) and activity (observed fold improvement) for Bxb1 integrase and found a weakly positive correlation between the two metrics, contrary to the other proteins reported evolves. This likely reflects a subset of protein families in which protein stability and fitness as learned by the PLM can predict activity as previously reported (67) (fig. S8C). However, given that the relationship is weak, a model such as EVOLVEpro is still needed to efficiently and quickly reach high-performing variants without encountering many false positives. Lastly, we found that the global mutation landscape learned by EVOLVEpro was still divergent from the predicted fitness (pMMS) by ESM2 with an even weaker correlation, further highlighting the ability of EVOLVEpro to learn protein activity at a global scale and how stability or fitness prediction is not sufficient for rapid and efficient protein evolution (fig. S8, D to G).

Evolving T7 RNA polymerase for efficient and highly pure RNA production

Multiobjective optimization with EVOLVEpro allows for the evolution of multiple complex activities. We chose to showcase multiobjective optimization on a common and broadly used enzyme with applications across basic biology and therapeutics. We selected the T7 RNA polymerase (RNAP) due to its critical role in RNA production for mRNA therapies, mRNA vaccines, cell engineering, and basic scientific studies. Because mRNA production has numerous features characterizing its potency and quality, as opposed to genome editing, where one feature matters the most, we designed a multiobjective optimization function to evolve a high-fidelity T7 RNAP for mRNA production with these three parameters: (i) RNA yield measured through ultraviolet-visible (UV-vis) spectrophotometry, (ii) mRNA translation in a double-stranded RNA (dsRNA) sensitive cell line measured by luciferase translation, and (iii) RNA purity measured through immunogenicity in BJ fibroblast cells by interferon β (IFN- β) RNA production (Fig. 5A). We weighted these features in the EVOLVEpro objective function by 20, 40, and 40%, respectively, to prioritize the higher fidelity and lower immunogenicity aspects of this enzyme for clinical applications. To facilitate high throughput variant testing, we relied on SP6 in vitro transcription-translation coupled reaction kits to generate mutant T7 RNAP in a one-pot reaction and subsequently use the produced T7 RNAP to produce cotranscriptionally capped Cypridina luciferase mRNA for downstream in vitro testing.

During the initial two rounds of evolution, fold improvement of top mutants was two- to fourfold. By rounds three and four, we started observing substantial improvements in all features, especially in translation and immunogenicity fold changes over the WT T7 RNAP (Fig. 5, B and C, and fig. S9A). By the end of round 4, one T7 RNAP mutant, Glu⁶⁴³Gly, generated luciferase mRNA that produced 34 $\times$ -more translated luciferase and ~98% less immunogenicity (Fig. 5C). We sought to benchmark Glu⁶⁴³Gly against the previously engineered state-of-the-art mutant T7 RNAP with Gly⁴⁷Ala mutation and ⁸⁸⁴Gly insertion (Gly⁴⁷Ala/⁸⁸⁴Gly) that has markedly reduced immunogenic byproduct in our in vitro transcription and translation (IVTT) assay (68). We found that our Glu⁶⁴³Gly mutant produces sevenfold-higher translation in cells and approximately

twofold less IFN- β 1 inflammation in BJ fibroblasts (fig. S9B).

To leverage the suite of mutants generated in the first four rounds, we generated multimutants with EVOLVEpro, combining up to three previously tested mutations. We also included combinations with the previously identified Gly⁴⁷Ala mutation known to reduce dsRNA formation. In typical rational mutagenesis, single beneficial mutations are combined according to their spatial location under the assumption of synergistic effects of these mutations. Here, we relied on EVOLVEpro's ability to learn the activity landscape to nominate multimutants. After two rounds of multimutants corresponding to the sixth round of engineering, EVOLVEpro-nominated variants with up to $\sim 57\times$ more translation from luciferase mRNA and $\sim 515\times$ less immunogenicity than the original WT T7 RNAP (Fig. 5C). The top variant, T7 RNAP^{Thr3Met/Gly47Ala/Glu643Gly}, was substantially more effective at translation and less immunogenic than the Gly⁴⁷Ala/^{884ins}Gly mutant. This multimutant was chosen as the final EVOLVEpro evolution candidate and termed EVOLVEpro-enhanced T7 RNAP, or epT7. The wide range of activities we see upon combining mutations points to complex epistatic interaction on the fitness landscape, and further examination of these interactions is needed to understand the mechanism underlying the range of activities seen for different combination mutants.

Given the high-throughput testing of mutant T7 RNAPs in the IVTT reaction, we hypothesized that the unoptimized IVT buffer could change these mutant's mRNA production (69) and sought to compare the performance of top mutants in clinically relevant IVT settings with NEB's HiScribe transcription kit, followed by Vaccinia cap-1 capping and poly-A tailing. We therefore purified the top performing single mutant (Glu⁶⁴³Gly), previously reported state-of-the-art mutant (Gly⁴⁷Ala/^{884ins}Gly) (68), and our epT7 (Thr³Met/Gly⁴⁷Ala/Glu⁶⁴³Gly) along with the WT to compare their performance. We compared the production of six different mRNA sequences, ranging in size from 500 to 6500 nt, between epT7, T7^{Glu643Gly}, and WT T7. Consistent with our IVTT-based experiment, we found that epT7 and Glu⁶⁴³Gly produced significantly higher mRNA in a 2-hour transcription scheme than both WT T7 RNAP and the Gly⁴⁷Ala/^{884ins}Gly variant (fig. S9C). Analysis of the three different mRNA products by both E-gel EX and TapeStation gel electrophoresis systems confirmed the presence of a single on-target product across all four enzymes (fig. S9, D to F). Looking at the translation and immunogenicity aspects of the mRNAs produced by these enzymes, we found that in all cases, epT7-produced mRNA had four- to 120-fold-higher translation than the WT and four- to 256-fold- lower immunogenicity (Fig. 5D

and fig. S10A). Functional testing of SpCas9 mRNA also shows significantly higher editing from epT7's produced mRNA in two separate cell lines (fig. S10B). These results validate that the EVOLVEpro-derived epT7 mutants are not buffer or template-specific and are genuinely improving the quality of mRNA produced by the polymerase. We next investigated the mechanism of the epT7 performance enhancements by investigating the quality of the RNA. By using an established ELISA for dsRNA, we found that the dsRNA in the epT7-produced mRNA was fivefold lower than WT T7-produced RNA, and it performed equally well as the RNA produced by the state-of-the-art Gly⁴⁷Ala/^{884ins}Gly mutant (68) (Fig. 5E).

Previous efforts to reduce dsRNA production relied on adding a glycine residue at the C-terminal "foot" region of the enzyme (⁸⁸⁴Gly insertion) (68). Our model revealed the functional importance of Glu⁶⁴³ in transcription, and, notably, mutating this residue rendered the same effect as ⁸⁸⁴Gly insertion (Fig. 5F and fig. S9B). Indeed, analysis of the T7 RNAP structure reveals that Glu⁶⁴³ is close to the DNA template, suggesting that Glu⁶⁴³Gly improves template binding and RNA production (Fig. 5F). However, Glu643Lys or Glu⁶⁴³Arg did not improve the fidelity of transcription (fig. S9A), suggesting that these bulky residues sterically clash with the template DNA.

To rationalize how EVOLVEpro is exploring the activity landscape, it is useful to consider the progression of nominated residues through the first four rounds. Glu⁶⁴³ was found first in round 3 with the most beneficial mutation being Glu⁶⁴³Asn (Fig. 5G). The model quickly zoomed into this region by exploring it five more times in round four, yielding Glu⁶⁴³Gly, the best single mutant. Gly⁴⁷Ala has been previously reported to increase helix formation, and EVOLVEpro took advantage of this helix-favoring mutation in our multimutant generation. The third mutated residue in epT7 is in a disordered region (T3M), suggesting a role independent of DNA template binding. T3M might be involved in improving protein stability or other aspects that can modulate the polymerase activities. These results suggest that EVOLVEpro can be used to identify and interrogate the effect of various mechanisms and determine the right balance biochemically to mutagenize.

We next calculated the relationship between the activity (observed data) and fitness (pMMS) for T7 RNAP and found a negative correlation of 0.13, in this case denoting the lack of association between the two metrics. EVOLVEpro successfully navigated through this divergence by selecting mutants with higher activity but not fitness in later rounds (Fig. 5H). Lastly, we investigated the global evolutionary landscape of epT7 and EVOLVEpro's mutational trajectory. At a high level, as with the previous proteins

evolved, the activity map learned by EVOLVEpro diverged from the fitness map predicted by ESM-2, showing that fitness predictions would not be able to predict the mutants that were ultimately discovered to improve protein activity and other parameters (Fig. 5, I to K, and fig. S10C).

Circular RNA production with epT7

Circular RNA has emerged as a promising therapeutic modality for protein replacement therapy thanks to its enhanced stability and prolonged expression of proteins (70). Because we observed significantly lower dsRNA production and higher fidelity of transcription with epT7, we hypothesized that epT7 would enhance circular RNA production, as the use of RNase R during post-IVT processing typically enriches for both circular RNA and dsRNA species that are immunogenic (Fig. 6A). We thus applied epT7 to the circularization of four different RNA sequences, finding that the translation obtained by circular RNA from epT7 is three- to 30-fold higher than RNA produced by WT T7 RNAP (Fig. 6B and fig. S11, A to D and J). We then used TapeStation gel electrophoresis to quantify the relative ratio of circular products after IVT and found reduced long concatemer formation in the circular RNA produced by epT7 (Fig. 6C). To better understand the mechanism behind better translation of circular RNA made by epT7, we performed gel electrophoresis using 2% E-gel EX, as previously validated, to check for the relative ratio of precursor, nicked, intermediate, and full circular RNA both pre- and post-RNase R treatment (Fig. 6D). We noticed reduced intermediate and nicked byproducts in circular RNA produced by epT7, showing higher fidelity of transcription. We used the gel electrophoresis results to quantify the ratio of circular RNA across three different templates and found significantly higher circular RNA production at around 25% efficiency, which was approximately twofold higher than the efficiency of WT T7 RNAP, higher circular RNA purity, and lower concatemer production (Fig. 6E and fig. S11, G to I). Lastly, we used dsRNA ELISA to detect the amount of dsRNA left in the product after RNase R cleanup. Consistent with our hypothesis, there was a large increase in dsRNA percentage at around 1.5% from WT T7's produced dsRNA (Fig. 6F). This dsRNA ratio was significantly reduced to 0.2% using epT7, highlighting the fidelity of this variant during the long transcription that is needed to accommodate circular RNA production (Fig. 6F). To confirm the higher stability of circular enhanced green fluorescent protein (eGFP) RNA, we transfected both WT T7 and epT7's produced circRNA in HEK293FT cells and imaged them 24 and 72 hours after transfection (fig. S11, E to F). We observed higher GFP fluorescence from epT7 than WT T7 RNAP and stable expression of GFP at 72 hours, similar to previously reported (70).

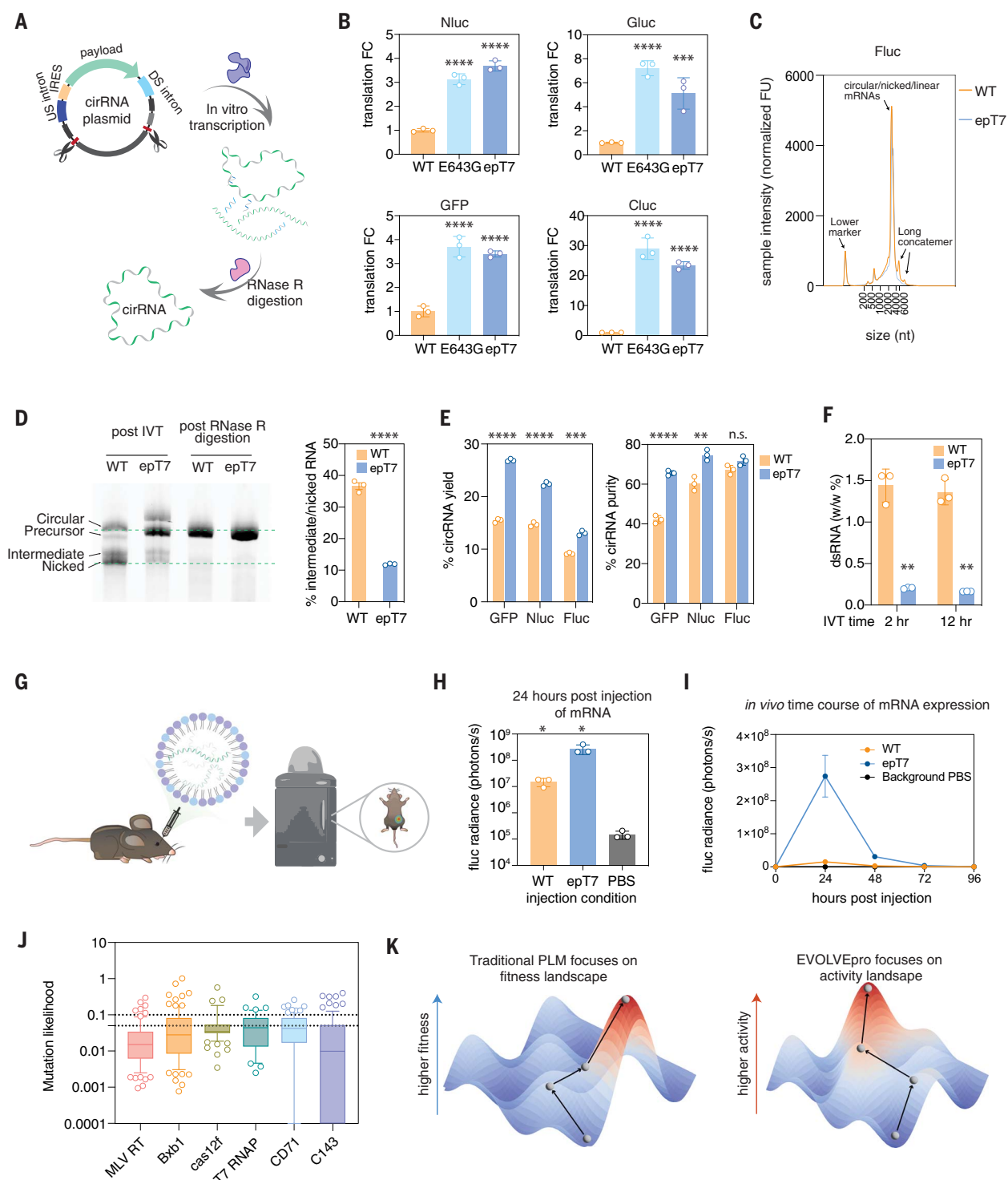


Fig. 6. Application of epT7 for circular RNA production and in vivo bioluminescence. (A) Schematic of circular RNA production. (B) Validation of epT7-produced circular RNA on four different template sequences compared to both WT T7 and WT T7. Translation of each protein is measured in HEK293FT cells 48 hours after transfection. A two-sided Student's *t* test was run between WT and each evolved T7 RNAP (***P* < 0.001; *****P* < 0.0001). Error bars represent standard deviation with *n* = 3 biological replicates. (C) TapeStation gel electrophoresis analysis of circular Fluc RNA produced by either epT7 or WT RNAP. epT7 shows reduced concatemer production. (D) Comparison of RNA products for Fluc circular RNA produced by epT7 compared

with WT T7 through gel electrophoresis using 2% E-gel EX at different steps in the production process: after initial IVT and after RNase R processing. (Right) The quantification of intermediate and nicked RNA ratio in the post IVT samples. Error bars represent standard deviation with *n* = 3 biological replicates. (E) Comparison of purified GFP, NanoLuc (Nluc), and Fluc circular RNA (circRNA) yield by epT7 compared to WT T7 after the initial RNase R cleanup. (Left) The raw mass percentage left after the cleanup. (Right) The purity of the circular RNA in the post clean-up reaction as determined by quantification using a TapeStation analysis. A two-sided Student's *t* test was run between WT and epT7 (***P* < 0.01; *****P* < 0.0001). (F) Comparison of dsRNA content for Nluc

circular RNA produced by epT7 compared to WT T7 using either 2 or 12 hours of IVT. Input into the dsRNA ELISA assay involves 500 ng of post-RNase R cleaned-up samples. A two-sided Student's *t* test was run between WT and evolved T7 RNAP (***P* < 0.01). Error bars represent standard deviation with *n* = 3 biological replicates. **(G)** Schematic of the in vivo mRNA assay for measuring mRNA expression in the liver through noninvasive luminescent imaging. **(H)** In vivo luminescent signal detected 24 hours after injection in mice injected with mRNA produced by either epT7 or WT T7 or PBS controls. A two-sided Student's *t* test was run between WT, WT T7 RNAP, and epT7 (**P* < 0.05). Error bars represent standard deviation with *n* = 3 biological replicates. **(I)** Time course of in vivo luminescent signal detected up to 96 hours after injection of LNP mRNA produced by PBS controls or either epT7 or WT T7. A two-sided paired Student's *t* test was run between WT, WT T7 RNAP, and epT7 (**P* < 0.05) for each time point. Error

bars represent the standard error of mean with *n* = 3 biological replicates.

(J) A box plot of mutational likelihood for each individual mutant nominated by EVOLVEpro shown for each of the six proteins in this study. The dashed lines at 0.05 and 0.1 are shown to denote the threshold for rare mutations and uncommon mutations, respectively. **(K)** A schematic showing the evolution of higher activity variants with EVOLVEpro versus traditional PLM evolution approaches. The mutagenesis landscape of proteins is often conceptualized as a complex terrain with numerous potential paths. Shown here is a gray road that conceptualizes the protein mutagenesis landscape, where traversing upwards results in higher protein activity, and traversing downwards reduces protein fitness. Traditional frameworks of evolutionary plausibility attempt to navigate this terrain based on natural selection, which is constrained by historical and environmental factors.

mRNA for in vivo bioluminescent imaging

Given the high fidelity of epT7, we compared the performance of epT7 with that of WT T7 RNAP in producing 100% N¹-methylpseudouridine-5'-triphosphate-modified firefly luciferase (Fluc) mRNA that is commonly used for in vivo deep tissue imaging (Fig. 6G). This production process, including the modified bases, mimics the clinical production of therapeutic mRNAs, allowing for a translationally relevant evaluation of epT7. We packaged the produced mRNA with lipid nanoparticles (LNPs) that traffic to the liver for bioluminescent imaging. At 24 hours after injection of the mRNA-loaded LNPs, we observed ~10-fold-higher luminescence for our epT7-produced mRNA compared with that of mRNA produced by WT T7 RNAP (Fig. 6H). Moreover, we tracked the expression kinetics of both mRNAs for 96 hours and found consistently higher translation with the epT7-produced Fluc mRNA for a longer period of time (Fig. 6I and fig. S11K).

Originality of mutations explored during EVOLVEpro evolution

We analyzed the mutations proposed by EVOLVEpro on the six proteins evolved in this study by calculating the mutational likelihood of an individual mutation compared with the training input (Uniprot). We found that the median mutational likelihood for each protein's set of mutations ranges from 0.01 to 0.04, which is well below the 0.05 cutoff for rare mutations (Fig. 6J) (15). Most of the mutations explored by the model are uncommon mutations not seen in nature as defined by a probability cutoff of <0.1, with 92% of PE2 MLV RT mutations and 77% of Bxb1 integrase mutations below this threshold. Moreover, all the best activity-enhancing single mutants explored during the evolution of the six proteins in this study have a mutational likelihood of <0.1. This analysis reveals that the mutation landscape explored by EVOLVEpro is highly original compared to zero shot-based language models and reinforces the need to search outside naturally occurring mutations to find activity-enhancing mutants (fig. S12).

Discussion

We demonstrate EVOLVEpro as a model for in silico-directed evolution of protein activities using few-shot active learning. Over consecutive rounds of improvement, EVOLVEpro yields variants with two- to 515-fold improvements in desired properties, including binding, catalytic efficiency, and immunogenic byproducts. By using both evolutionary-scale PLMs and a regression layer, EVOLVEpro learns general rules of protein activity, generating highly active mutants with only a few cycles of evolution. Moreover, because of the rich latent space generated by the PLM and powerful feature selections present in the top-layer module, EVOLVEpro evolution is a low-N learning approach that requires minimal wet lab experimentation. We benchmarked EVOLVEpro across 12 different DMS datasets covering eight protein classes, showing its superiority in the low-N evolution setting. In this benchmarking work, we evaluate all currently available embedding-based PLMs and perform a grid search to optimize over top-layer regression models, active learning selection strategies, and different normalization techniques toward the embeddings and activity measurements. We found that PLMs are essential, and their representations of protein sequence outperform traditional encoding methods, such as one-hot encoding and integer encoding (Fig. 1B). Notably, even in the extreme scarcity of data relative to the size of the input vector, dimensionality reduction of the embedding space through PCA did not improve performance, reinforcing the importance of the PLM dimensions in guided in silico-directed evolution (see methods and data S1). The modular design of EVOLVEpro allows for the integration of future improvements in autoregressive PLMs or next-generation representation models.

The success of EVOLVEpro speaks to the inherent limitations of PLMs, which are trained to learn a masked sequence reconstruction task across evolutionary diversity. Because natural sequences do not necessarily select for optimal protein activity, the PLM's learned activity landscape will often not be correlated with a protein's activity landscape (Fig. 6K). In scenarios of cor-

relations between fitness and activity, such as antibodies, zero-shot PLM protein evolution may work with some success (15, 20), but enzyme optimization has proven more challenging. It has been shown that PLMs can scale with increasing parameters just like large language models, but recent analyses have shown saturating scaling effects of PLMs with limited input training datasets (Uniref) on larger models (71–74). Thus, it is likely that simply increasing the parameters of these PLMs will not enable better prediction of protein activities and other downstream tasks. Alternatively, generative PLMs have yielded functional de novo proteins, such as GFP and CRISPR nucleases (16, 19). These models explore a much larger search space than EVOLVEpro in the initial design phase, but variant designs generated by these methods do not have improved activities relative to WT proteins yet, and the functional success rate of generated proteins is very low. As such, Rufollo *et al.* successfully designed OpenCRISPR, an artificial intelligence (AI)-generated Cas9 protein that has cleavage efficiency comparable with that of the WT SpCas9 (19). We expect generative PLMs to design fairly active protein sequences in tasks where there doesn't exist a good starting point (i.e., a binder against a previously unknown target) (75). These de novo-designed sequences may be suitable for combination with EVOLVEpro to create an end-to-end de novo design and evolution framework, where de novo-generated sequences can be rapidly optimized for state-of-the-art activity and, thus, real-world deployment. In addition, biophysical-based models can also be integrated with the regressive top-layer approach established in this work to further boost prediction accuracy and enable rapid identification of gain-of-function mutants in silico (76).

By using EVOLVEpro, we present the first comprehensive evaluation of an AI-directed evolution model across six therapeutically relevant proteins. These proteins demonstrate a low correlation between observed activity and the PLM-estimated fitness, requiring EVOLVEpro to rapidly navigate the unseen activity landscape. In some cases, we leverage EVOLVEpro for multiobjective feature optimization, allowing

evolution of multiple properties simultaneously. Critically, the assays used for measuring protein activity in this work are incompatible with pooled screening approaches, precluding typical directed evolution strategies. Across the multimutant landscape of protein activity, EVOLVEpro is able to select highly active single mutants out of >16,000 possible sequences and multimutants from more than 780 billion possible sequences. We thoroughly validated the six proteins evolved by EVOLVEpro for genome editing, binding, and RNA generation tasks beyond the training set, finding state-of-the-art performance. Structural analysis of top mutations reveals many distinct mechanisms of activity improvement, suggesting future directions for directed evolution of these enzymes. In the context of protein design, EVOLVEpro is a highly capable protein engineering model in that it (i) has high rates of success, (ii) requires no special knowledge about the protein, (iii) can be used for multi-objective function or property optimization, and (iv) is highly modular, allowing for any protein property with a quantifiable assay to be used as an input without extensive finetuning. We anticipate that EVOLVEpro will continue to improve with new foundation models and enhanced search strategies and will be broadly useful for protein engineering.

Methods summary

Use of ESM2 embeddings

Let $x_i = [a_1, a_2, \dots, a_n]$ denote the amino acid sequence of the i -th protein variant, where each a_j represents an individual amino acid and n is the length of the protein. The protein language model embedding transformation (ESM2-15B) maps x_i to a sequence of embeddings, one for each amino acid, where d is the dimensionality of the embedding space (hidden dimension):

$$E_i = \text{PLM}(x_i) \in \mathbb{R}^{n \times d} \quad (1)$$

This results in a per-token representation of $\text{size} \times d$. We use the final representation layer of the ESM2-15B model. To reduce the number of features in the low- N setting and obtain a fixed-size representation regardless of protein length, we compute the average embedding vector by taking the mean across all amino acid positions:

$$\bar{e}_i = \frac{1}{n} \sum_{j=1}^n E_{ij} \quad (2)$$

This results in a single d -dimensional vector, $\bar{e}_i \in \mathbb{R}^d$, representing the entire protein variant, which is then used as input for EVOLVEpro.

EVOLVEpro model

EVOLVEpro utilizes a Random Forest regressor as its top-layer model to learn the functional grammar of variants with respect to their activity. This model operates on information-

rich latent space mean embeddings $\bar{e}_i$ generated by a protein language model as described previously, in an active learning setting. The Random Forest regressor uses these embeddings $\bar{e}_i$ as input features to predict the activity or fitness of each variant.

A Random Forest Regressor was employed as the top-layer model in the EVOLVEpro framework. This ensemble learning method combines multiple decision trees to make predictions, offering robustness against overfitting and the ability to capture complex, nonlinear relationships in the data.

The Random Forest model was configured with 100 estimators (individual decision trees). The quality of splits was evaluated using the Friedman Mean Squared Error (MSE) criterion.

Let $D = \{(\bar{e}_i, y_i)\}_{i=1}^N$ be the training dataset, where $\bar{e}_i \in \mathbb{R}^d$ are the input features (reduced embeddings), and $y_i \in \mathbb{R}$ are the target values (protein fitness).

Decision trees

Each tree $h_t(\bar{e}_i)$ in the forest is trained on a bootstrap sample of the original dataset. At each node, the best split is determined by maximizing the reduction in impurity:

$$\Delta I = I(\text{parent}) - \frac{N_{\text{left}}}{N} I(\text{left}) - \frac{N_{\text{right}}}{N} I(\text{right}) \quad (3)$$

where I is the Friedman MSE impurity measure, calculated as follows:

$$\text{diff} = \bar{y}_{\text{left}} - \bar{y}_{\text{right}} I = \frac{N_{\text{left}} \cdot N_{\text{right}}}{N_{\text{left}} + N_{\text{right}}} \cdot \text{diff}^2 \quad (4)$$

Here, $\bar{y}_{\text{left}}$ and $\bar{y}_{\text{right}}$ are the mean target values in the left and right child nodes, respectively, and N_{left} and N_{right} are the number of samples in each child node.

Random Forest prediction

The final prediction of the Random Forest for a new input $\bar{e}_i$ is the average of the predictions from all trees:

$$f(\bar{e}_i) = \frac{1}{T} \sum_{t=1}^T h_t(\bar{e}_i) \quad (5)$$

where $T = 100$ is the number of trees in the forest.

Active learning approach

For each round of EVOLVEpro, after training the Random Forest on the current dataset, we apply the model to predict fitness values for the remaining, untested protein variants. The active learning step then selects the most promising variants for experimental testing in the next round.

Given the nature of Random Forest regression, all predicted values for untested variants

will fall within the range of y -values in the training set:

$$\min(y_{\text{train}}) \leq \hat{y}_{\text{predict}} \leq \max(y_{\text{train}}) \quad (6)$$

where y_{train} is the fitness values in the training set, and $\hat{y}_{\text{predict}}$ is the predicted fitness values for untested variants.

Although we explored various selection strategies, including choosing embeddings most distant from the training set in Euclidean space and selecting a mix of top and bottom predictions, we hypothesize that for the objective of round-over-round optimization of top fitness, the top- n strategy is most effective.

The top- n strategy involves selecting the n variants with the highest predicted fitness values:

$$\text{Selected Variants} = \{\bar{e}_i : \hat{y}_i \geq y_{(n)}\} \quad (7)$$

where $y_{(n)}$ is the n th highest predicted fitness value.

This strategy aims to iteratively stretch the upper limit of the Random Forest's prediction range:

$$\max(y_{\text{train}})_{\text{round } k} \leq \max(y_{\text{train}})_{\text{round } k+1} \quad (8)$$

By focusing on the top predictions, we exploit the model's current understanding of high-fitness regions in the embedding space, while also encouraging exploration of nearby areas that may yield even better variants. Although this greedy approach might risk overlooking some areas of the fitness landscape, we believe it aligns well with the goal of rapidly identifying and optimizing top-performing protein variants in a limited number of experimental rounds.

Benchmarking on 12 DMS datasets

For model benchmarking, we took nine existing DMS datasets which were employed in a previous zero-shot high fitness prediction approach (15). From this work, we leveraged a predetermined cutoff for high-fitness variants for each dataset to select variants that were low and high fitness. To augment the use of these datasets, we also selected three additional DMS datasets: an AsCasI2f compact genome editor (33), Cov2 viral spike receptor-binding domain (43), and Zika virus envelope protein (42). For these datasets, cutoff values were set based on the general distribution of high-activity variants. To facilitate downstream work, a tabular format CSV file and a fasta file of all available mutant sequences with activity measurements were generated from each dataset.

EVOLVEpro parameter grid search

We conducted an extensive grid search to evaluate various strategies for optimizing fitness in a low number of rounds. The grid search explored the following parameters:

1) Fitness measurement: Raw fitness values from each dataset or min-max normalized fitness.

2) First-round strategy: Random selection of variants or diverse selection using K-medoids clustering on protein language model embeddings.

3) Learning strategies: We compared several strategies for selecting variants in subsequent rounds, including:

- a) Random selection.
- b) Top n predicted fitness variants.
- c) Top n/2 and bottom n/2 predicted fitness variants.

d) Maximizing Euclidean embedding distance from previously selected variants.

4) Embedding types: We compared different embedding representations, including raw embeddings, and PCA-reduced embeddings (from 10 to 1000) to account for the fact that this was a high-p, low-n paradigm. This was entirely done on the largest (15B parameter) ESM2 model.

5) Regression types: We evaluated various regression models for fitness prediction, including ridge regression, lasso regression, elastic net, linear regression, neural networks with a linear last layer, random forest regression, KNN regression, Gaussian processes, and gradient boosting regression. These were largely used with default parameters.

For each combination of parameters, we ran three simulations (to vary the first round of selected variants) using 16 variants per round to account for stochastic variability. Performance was assessed using the proportion of high-fitness variants out of the top 16 variants that the updated model would predict. We quantified the overall effectiveness of each parameter value by counting the number of datasets for which it achieved the highest mean fitness binary percentage. This “winning strategy” count provided a simple yet informative summary of which approaches were most successful across diverse protein systems. The “winning strategy” was random first round, raw fitness, top-n selection, random forest regression, and raw embeddings.

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SUPPLEMENTARY MATERIALS

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